
AnomalyClaw: Adversarial VLM Debate with Expert Grounding for Cross-Domain Visual Anomaly Detection

Anonymous Authors

Abstract

Vision-language models (VLMs) offer strong semantic reasoning for visual anomaly detection (VAD), yet they lack quantitative grounding, hallucinate defects on unfamiliar images, and cannot self-verify their judgments. We present ANOMALYCLAW, a training-free framework that detects anomalies through *adversarial debate* between two VLM-based agents—an Anomaly Proposer that actively searches for defects and a Normality Advocate that challenges each claim with counter-evidence—grounded by an extensible pool of lightweight vision experts. An autonomous controller orchestrates the process: it selects which experts to invoke based on the query context, feeds their structured evidence into the debate, and terminates when claims converge or a computational budget is exhausted. This design replaces rigid pipelines with a self-regulating reasoning loop where the VLM decides what evidence it needs and when it has seen enough. We evaluate on a cross-domain benchmark spanning 12 diverse domains (industrial, 3D, remote sensing, medical, road-safety, and logical anomalies) with 1,440 test items. AnomalyClaw achieves **X.XXX** macro-averaged AUROC, improving over single-pass VLM baselines (**X.XXX**) and prior agent-based methods (**X.XXX**). Ablations confirm that adversarial debate outperforms single-agent reasoning by **+X.X** AUROC points, and that expert grounding contributes an additional **+X.X** points. The framework generalizes across multiple VLM backends without modification. Code, benchmark, and evaluation scripts will be released upon acceptance.

1 Introduction

Visual anomaly detection (VAD) is a prerequisite for safety in manufacturing, medical imaging, infrastructure inspection, and autonomous driving. Recent work has explored using vision-language models (VLMs) for anomaly detection, leveraging their strong semantic reasoning capabilities [Gu et al., 2024, Jiang et al., 2025, Chen et al., 2026]. Yet three fundamental limitations prevent VLMs from serving as reliable cross-domain anomaly detectors.

Three VLM limitations in VAD. (1) *Lack of quantitative grounding.* VLMs reason about visual content in natural language but cannot compute patch-level distance statistics that make expert detectors like PatchCore [Roth et al., 2022] effective. When a subtle surface scratch changes patch distances by 0.15 standard deviations, the VLM sees only pixels and may miss it entirely. (2) *Hallucination and under-detection.* Without quantitative anchoring, VLMs hallucinate defects on unfamiliar but normal images or fail to detect genuine anomalies when they lack domain-specific visual priors [Jiang et al., 2025]. (3) *No self-verification.* A single-pass VLM judgment has no mechanism to catch its own errors. The model cannot revisit a claim when contradictory evidence emerges, nor can it request additional expert analysis when its initial assessment is uncertain.

Existing approaches address these limitations in isolation. *Vision-expert* methods [Roth et al., 2022, Oquab et al., 2024, Jeong et al., 2023] provide precise patch-level scores but cannot handle se-

AnomaClaw — 3-Stage Visual Anomaly Detection Pipeline

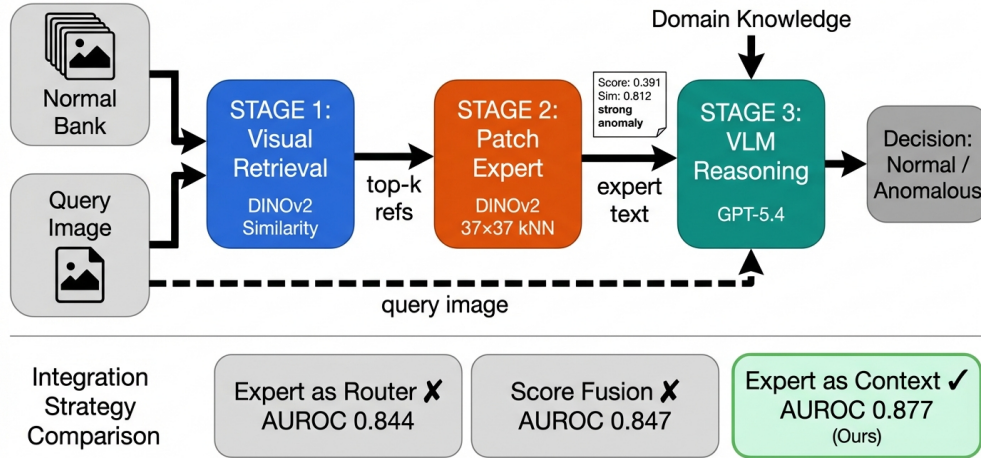


Figure 1: **AnomalyClaw overview.** An extensible expert pool provides quantitative evidence (patch distances, retrieval similarity, texture statistics) as structured text. The Anomaly Proposer uses this evidence to identify potential defects with spatial locations and confidence scores. The Normality Advocate challenges each claim with counter-evidence (lighting variation, normal texture, manufacturing tolerance). An autonomous controller decides which experts to invoke, when to launch additional debate rounds, and when the system has converged.

mantic or logical anomalies. *VLM-only* methods [Gu et al., 2024, Jiang et al., 2025, Cao et al., 2025] reason fluently but lack quantitative grounding and self-correction. Recent tool-augmented approaches [Miao et al., 2025, Chen et al., 2026, Ji et al., 2025] combine VLMs with expert tools, but use fixed pipelines where the integration strategy is predetermined—the system cannot adapt its reasoning depth or expert selection based on the difficulty of each query.

Our approach: adversarial debate with expert grounding. AnomalyClaw addresses all three limitations through a principled framework built on three components (Figure 1).

First, an **extensible expert pool** provides quantitative grounding. Rather than hardcoding a single expert (e.g., PatchCore), the framework supports a collection of lightweight analysis modules—patch-level nearest-neighbor scoring, visual retrieval, texture statistics, and others—each producing structured text evidence. The expert pool is open: new experts can be added without modifying the debate engine.

Second, an **adversarial debate engine** enables self-verification. An *Anomaly Proposer* examines the query image alongside expert evidence and proposes specific anomaly claims—each with a spatial location, appearance description, and confidence level. A *Normality Advocate* then challenges each claim, arguing that the observation is a normal variation (lighting, texture, manufacturing tolerance) rather than a genuine anomaly. Claims that survive refutation are accepted; those successfully challenged are rejected; ambiguous claims trigger additional debate rounds. This adversarial structure forces the system to justify every detection against explicit counter-arguments, directly addressing the hallucination and self-verification problems.

Third, an **autonomous controller** orchestrates the process. It determines which experts to invoke based on the initial assessment, controls debate depth (easy cases converge in one round; hard cases may require multiple), and manages the computational budget. The controller replaces hand-crafted decision rules (e.g., family-specific fusion weights) with a self-regulating loop where the system dynamically allocates effort based on query difficulty.

Contributions.

1. **Adversarial debate for anomaly detection.** We introduce an adversarial reasoning framework where an Anomaly Proposer and a Normality Advocate debate over each query image, grounded by expert evidence. This design provides built-in self-verification: every anomaly claim must survive explicit counter-arguments before being accepted. Ablations show that adversarial debate outperforms single-agent reasoning by **+X.X** AUROC points.
2. **Extensible expert pool with autonomous invocation.** Expert evidence is provided through a modular pool of lightweight analysis modules (patch-level scoring, visual retrieval, texture analysis, etc.), selected and orchestrated by an autonomous controller that adapts to query difficulty. This replaces rigid pipeline designs with a self-regulating reasoning loop.
3. **Cross-domain VAD benchmark.** We assemble a unified evaluation spanning 12 domains—industrial manufacturing (MVTec-AD), retail products (GoodsAD), complex and multi-object industrial inspection (VisA), infrastructure (SDNET2018), logical anomaly (MVTec-LOCO), 3D product inspection (Real3D-AD), satellite change detection (LEVIR-CD+), dermatology, brain MRI, liver CT, GI endoscopy, and road safety—comprising 1,440 test items with standardized metrics and protocols.
4. **VLM-agnostic generalization.** We demonstrate that AnomalyClaw works across multiple VLM backends (**GPT-4o**, **GPT-5.4**, **open-source VLMs**) without modification, achieving consistent improvements over single-pass baselines regardless of the underlying model.

The remainder of the paper is organized as follows. Section 2 reviews VLM-based and expert-based anomaly detection. Section 3 details the AnomalyClaw architecture. Section 4 presents experiments and ablations, and Section 5 concludes.

2 Related Work

Training-Free Visual Anomaly Detection. A growing body of work tackles anomaly detection without task-specific training by leveraging powerful pretrained representations. PatchCore [Roth et al., 2022] introduced a coreset-based memory bank of patch-level features, enabling efficient nearest-neighbor anomaly scoring that remains competitive on industrial benchmarks. WinCLIP [Jeong et al., 2023] extended this to the zero-shot setting using CLIP-based window-level similarity scores. Self-supervised vision transformers—particularly DINOv2 [Oquab et al., 2024]—have provided rich patch features that further improve nearest-neighbor methods, and FoundAD [Li, 2026] showed that feature adaptation strategies can close the gap between training-free and fine-tuned approaches. Despite strong performance on MVTec and VisA, these methods have been evaluated almost exclusively on industrial surfaces; their generalization across medical, retail, road, and logical anomaly domains remains unexplored. AnomalyClaw incorporates patch-level experts as one component of a broader expert pool, rather than relying on a single detector as the sole decision-maker.

VLM-Based Anomaly Detection. Vision-language models offer semantic reasoning that purely visual detectors lack. AnomalyGPT [Gu et al., 2024] fine-tunes a VLM to produce anomaly judgments and localization maps. Anomaly-OneVision [Cao et al., 2025] distills anomaly knowledge into a unified VLM handling multiple industrial categories. The MMAD benchmark [Jiang et al., 2025] formalized VLM-based anomaly detection as visual question answering, revealing that even state-of-the-art VLMs struggle with fine-grained defect recognition. AD-Copilot [Zhang, 2026] proposed an interactive assistant paradigm. A common limitation is the absence of quantitative grounding and self-verification: VLMs make single-pass judgments with no mechanism to catch their own errors. AnomalyClaw addresses both gaps through adversarial debate grounded by structured expert evidence.

Agent-Based Anomaly Detection. Tool-augmented agents have been applied to compensate for VLM perceptual weaknesses. AgentIAD [Miao et al., 2025] equips a VLM with external tools for feature extraction and comparison, training the agent via supervised fine-tuning. EAGLE [Chen et al., 2026] injects PatchCore-derived visual prompts (heatmaps, bounding boxes) directly into the VLM’s visual input. AutoIAD [Ji et al., 2025] coordinates multiple specialized agents to automate the detection pipeline. These systems demonstrate the value of combining VLMs with specialized detectors but use *fixed* integration strategies: the order and type of tool invocation is predetermined.

Table 1: **Comparison with VLM-based methods for anomaly detection.** AnomalyClaw is the only system that (i) uses adversarial debate for self-verification, (ii) supports an extensible expert pool with autonomous invocation, (iii) requires no training, and (iv) is evaluated across 10 diverse domains.

Method	Expert	Channel	Self-Verify	Training	Adaptive	Cross-Domain
AgentIAD 2025	PatchCore+CLIP	Tool calls	No	SFT	No	No
EAGLE 2026	PatchCore	Visual overlay	No	None	No	Partial
AutoIAD 2025	Multi-agent	Pipeline	No	None	No	No
DMAD 2026	None	Debate	Partial	None	No	No
AnomalyClaw	Extensible pool	Text context	Yes	None	Yes	Yes (10)

regardless of query difficulty. AnomalyClaw differs by making expert selection and reasoning depth *adaptive*: the autonomous controller decides which experts to invoke and how many debate rounds to run based on the evolving state of the debate. Table 1 summarizes these differences.

Multi-Agent Debate and Reasoning. Multi-agent debate—where multiple LLM instances critique and refine each other’s answers—has shown promise for improving factual accuracy. DMAD [Wang and Li, 2026] applied multi-agent deliberation to anomaly detection through iterative discussion among VLM agents. M3MAD-Bench [Liang, 2026] studied multi-agent debate for multimodal tasks, identifying “collective delusion” as a key failure mode where agents reinforce rather than correct errors. VipAct [Zhang et al., 2026] proposed a vision-perception-action framework with role-based collaboration. These systems use *symmetric* debate where both agents play identical roles. AnomalyClaw uses *asymmetric* debate with structurally different roles (Proposer vs. Advocate), which avoids collective delusion by design: the Advocate’s sole objective is to challenge anomaly claims, preventing the echo-chamber effect observed in symmetric debate [Liang, 2026]. Furthermore, AnomalyClaw grounds the debate with quantitative expert evidence rather than relying solely on VLM perception.

Cross-Domain Anomaly Detection Benchmarks. MMAD [Jiang et al., 2025] covers industrial defect detection across MVTec-AD and VisA but does not extend beyond manufactured objects. ADNet [Ling et al., 2025] and M3-AD [Chen, 2025] broaden coverage to additional industrial categories. BMAD [Bao et al., 2024] addresses medical domains but does not consider non-medical settings. Our evaluation framework curates tasks from 10 domains under one unified protocol, enabling a broader study of how VLM-based detectors generalize across fundamentally different notions of visual anomaly.

3 Method

We present AnomalyClaw, a training-free framework for cross-domain visual anomaly detection built on three components: an *extensible expert pool* that provides quantitative grounding, an *adversarial debate engine* that enables self-verification through structured argumentation, and an *autonomous controller* that orchestrates expert selection and debate depth.

3.1 Problem Formulation

Given a query image I^q , a domain code d identifying the inspection context (e.g., `mvtec_metal_nut`, `brain_mri`), and a bank of normal images $\mathcal{B}_d = \{I_1^n, \dots, I_N^n\}$, the goal is to produce an anomaly verdict $v \in \{\text{normal}, \text{anomalous}\}$ and a set of anomaly claims with confidence scores. The system operates in a *training-free, few-shot* setting: no parameters are learned from anomaly examples, and only k normal references (typically $k \leq 5$) are used per query at inference time.

3.2 Extensible Expert Pool

The expert pool $\Phi = \{\phi_1, \dots, \phi_E\}$ is a collection of lightweight analysis modules, each producing structured text evidence about the query image. Any expert that takes an image (and optionally

reference images) as input and produces a textual report can be integrated without modifying the debate engine.

Patch-level expert. Inspired by PatchCore [Roth et al., 2022], this expert extracts dense patch tokens using DINOv2 [Oquab et al., 2024] at 518×518 resolution (37×37 grid, $M = 1,369$ patches per image). For each query patch $\mathbf{p}_i^q$, we compute the cosine distance to its nearest neighbor in the reference patch bank $\mathcal{P}_{\text{ref}} = \bigcup_{I^r \in \mathcal{R}} \{\mathbf{p}_1^r, \dots, \mathbf{p}_M^r\}$:

$$d_i = 1 - \max_{\mathbf{p} \in \mathcal{P}_{\text{ref}}} \cos(\mathbf{p}_i^q, \mathbf{p}). \quad (1)$$

The expert anomaly score aggregates the top-1% most anomalous distances:

$$s_{\text{patch}} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} d_i, \quad \mathcal{T} = \{i : d_i \geq d_{(0.99M)}\}. \quad (2)$$

The score is converted to a structured text report including the raw distance, global CLS-token similarity, and a qualitative assessment (“very similar to normal” through “strong anomaly signal”).

Visual retrieval expert. This module retrieves the k most similar normal images from $\mathcal{B}_d$ using DINOv2 CLS-token cosine similarity and reports the retrieval scores, providing a global similarity context that complements the patch-level analysis.

Additional experts (extensible). The pool can include texture-statistics experts (Gram matrices, frequency-domain analysis), CLIP-based semantic similarity, or domain-specific modules (e.g., Hounsfield-unit windowing for CT images). Each expert follows the same interface: accept image and references, return structured text. The autonomous controller (Section 3.4) decides which experts to invoke per query.

3.3 Adversarial Debate Engine

The core of AnomalyClaw is a structured debate between two VLM-based agents with opposing objectives (Figure 1).

Anomaly Proposer ($\mathcal{P}$). Given the query image, reference images, domain knowledge, and expert evidence, the Proposer generates a set of anomaly claims $\mathcal{C} = \{c_1, \dots, c_n\}$. Each claim is a structured record:

$$c_j = (\text{id}_j, \text{location}_j, \text{appearance}_j, \text{evidence}_j, c_j, \text{severity}_j), \quad (3)$$

where location_j specifies the spatial region (bounding box or quadrant), appearance_j describes the visual observation, evidence_j links the claim to specific expert findings, $c_j \in [0, 1]$ is the Proposer’s confidence, and severity_j is the assessed impact. If no anomalies are found, the Proposer outputs a normal profile describing the expected appearance of the object.

Normality Advocate ($\mathcal{A}$). For each claim c_j , the Advocate generates a counter-argument:

$$a_j = (\text{id}_j, r_j, \text{counter_evidence}_j, \text{likely_cause}_j), \quad (4)$$

where $r_j \in [0, 1]$ is the Advocate’s confidence that the observation is *not* anomalous, $\text{counter_evidence}_j$ provides specific reasons (“lighting reflection matches reference angle,” “texture variation is within normal manufacturing tolerance”), and likely_cause_j attributes the observation to a benign source. The Advocate’s sole objective is to argue against anomaly claims, ensuring that only robust claims survive.

Claim resolution. After each debate round, claims are resolved by comparing the Proposer’s confidence against the Advocate’s refutation confidence:

$$\text{status}(c_j) = \begin{cases} \text{Valid} & \text{if } c_j \geq \tau_{\text{accept}} \text{ and } r_j \leq \tau_{\text{reject}}, \\ \text{Invalid} & \text{if } r_j \geq \tau_{\text{accept}}, \\ \text{TBD} & \text{otherwise,} \end{cases} \quad (5)$$

where $\tau_{\text{accept}} = 0.6$ and $\tau_{\text{reject}} = 0.4$ are resolution thresholds. Claims with status TBD are carried into the next debate round for re-examination.

Algorithm 1 AnomalyClaw adversarial debate

Require: Query image I^q , domain d , normal bank $\mathcal{B}_d$, expert pool Φ , max depth D

Ensure: Verdict v , claim decisions

```
1: // Expert Evidence Collection
2:  $\mathcal{E} \leftarrow \Omega.\text{SelectExperts}(I^q, d, \Phi)$  {Autonomous expert selection}
3:  $\text{reports} \leftarrow \{\phi(I^q, \mathcal{R}, \mathcal{B}_d) : \phi \in \mathcal{E}\}$ 
4: // Debate Initialization (Round 0)
5:  $\mathcal{C} \leftarrow \mathcal{P}(I^q, \mathcal{R}, \text{reports}, d)$  {Propose anomaly claims}
6:  $\text{rebuttals} \leftarrow \mathcal{A}(I^q, \mathcal{R}, \mathcal{C})$  {Challenge claims}
7:  $\text{decisions} \leftarrow \text{Resolve}(\mathcal{C}, \text{rebuttals})$  {Eq. 5}
8: // Multi-Round Refinement
9: for  $t = 1, \dots, D - 1$  do
10:    $\text{TBD} \leftarrow \{c_j : \text{decisions}[j] = \text{TBD}\}$ 
11:   if  $\text{TBD} = \emptyset$  then
12:     break {All claims resolved}
13:   end if
14:    $\mathcal{C}' \leftarrow \mathcal{P}.\text{Refine}(\text{TBD}, \text{rebuttals})$ 
15:    $\text{rebuttals}' \leftarrow \mathcal{A}.\text{Refine}(\text{TBD}, \mathcal{C}')$ 
16:    $\text{decisions} \leftarrow \text{Resolve}(\mathcal{C}', \text{rebuttals}')$ 
17: end for
18: // Final Verdict
19:  $v \leftarrow \Omega.\text{Verdict}(\text{decisions})$ 
20: return  $v$ , decisions
```

Multi-round refinement. When TBD claims remain after the first round, the Proposer and Advocate engage in additional rounds focused exclusively on unresolved claims. The Proposer may revise confidence levels based on the Advocate’s counter-arguments, and the Advocate may strengthen or weaken its challenges based on new evidence. The debate terminates when all claims reach *Valid* or *Invalid* status, or when the maximum depth D (controlled by the autonomous controller) is reached.

Why asymmetric debate avoids collective delusion. Symmetric multi-agent debate [Wang and Li, 2026, Liang, 2026] assigns identical roles to all agents, which can lead to “collective delusion”—agents reinforcing each other’s errors through echo-chamber dynamics. AnomalyClaw’s asymmetric design avoids this by construction. The Advocate’s *sole objective* is to argue against anomaly claims; it succeeds by finding convincing counter-evidence, not by agreeing with the Proposer. This structural asymmetry ensures that false positives face genuine resistance, while true anomalies that survive the Advocate’s challenges are accepted with higher confidence than single-pass judgments.

3.4 Autonomous Controller

The autonomous controller Ω orchestrates the debate process, making three decisions:

Expert selection. Not all experts are informative for every query. The controller first runs a lightweight assessment (e.g., global CLS-token similarity from the retrieval expert) and selects additional experts based on the initial signal:

- If global similarity is high (query closely matches references), the patch-level expert is likely unnecessary for easy-normal cases.
- If global similarity is ambiguous, all available experts are invoked to provide maximum evidence.
- Domain-specific experts (e.g., CT windowing) are invoked only for relevant domains.

This adaptive selection reduces computational cost on easy queries while providing comprehensive evidence for difficult ones.

Debate depth control. The controller sets the maximum debate depth D based on the initial assessment:

$$D = \begin{cases} 1 & \text{if initial confidence} \geq \tau_{\text{easy}} \text{ (clear case),} \\ D_{\text{max}} & \text{otherwise (ambiguous case),} \end{cases} \quad (6)$$

where τ_{easy} is a confidence threshold (typically 0.8) and D_{max} is the maximum allowed depth (typically 2–3). Easy cases—where the Proposer and Advocate quickly agree—terminate in a single round, saving the cost of additional VLM calls.

Final verdict. The controller produces the final anomaly verdict from the debate outcomes:

$$v = \begin{cases} \text{anomalous} & \text{if any claim has status Valid,} \\ \text{normal} & \text{if all claims are Invalid or no claims proposed,} \\ \text{uncertain} & \text{if TBD claims remain at depth } D. \end{cases} \quad (7)$$

Uncertain cases can be routed to human review or flagged for additional analysis, providing a natural interface for human-in-the-loop deployment.

3.5 VLM-Agnostic Design

AnomalyClaw is designed to work with any VLM that supports multi-image input and structured (JSON) output. The Proposer and Advocate communicate through structured JSON schemas (Equations (3) and (4)), and expert evidence is provided as plain text. This design avoids dependence on any specific model’s capabilities or API format. In our experiments (Section 4), we evaluate with **multiple VLM backends** and show consistent improvements across all of them.

4 Experiments

We evaluate AnomalyClaw on a 12-domain benchmark spanning industrial, 3D, remote sensing, medical, scene-level, and logical anomaly detection. All code and evaluation scripts will be released upon publication.

4.1 Experimental Setup

Benchmark. We construct a cross-domain evaluation suite (Table 2). Each domain contributes 120 test images (60 normal, 60 anomalous) drawn from established public datasets, totaling 1,440 test items. Domains are organized sequentially by type: industrial (D1–D4), logical (D5), 3D point cloud (D6), remote sensing (D7), medical (D8–D11), and scene (D12). VisA (D3) is included specifically for its multi-object categories (PCBs, capsules) and complex texture objects. Real3D-AD (D6) provides rendered point cloud views testing whether VLMs can detect geometric deformations. LEVIR-CD+ (D7) introduces satellite imagery change detection, testing cross-modality generalization.

Baselines. We compare against methods spanning three categories:

Expert-only methods (no VLM):

1. **CLIP-ZeroShot:** Zero-shot anomaly classification using CLIP text-image similarity with prompt templates (“a photo of a normal/defective [object]”).
2. **WinCLIP** [Jeong et al., 2023]: Window-level CLIP similarity with multi-scale aggregation.
3. **PatchCore Expert:** Our patch-level expert in isolation (DINOv2 features, top-1% aggregation, $k=4$ retrieved references).

Single-pass VLM methods (one VLM call, no debate):

4. **VLM-Direct:** VLM receives query image only, no references or expert evidence.
5. **Retrieval+VLM:** DINOv2 retrieval followed by VLM comparison between query and references.

Table 2: **Cross-domain benchmark overview.** Twelve domains organized by anomaly type. Each domain contributes 120 test images (60 normal, 60 anomalous).

Code	Domain	Source	Type	Challenge
D1	Industrial Mfg.	MVTec-AD	Local	Texture defects
D2	Retail Products	GoodsAD	Local	Packaging damage
D3	Complex Industrial	VisA	Local+Multi	Multi-object, complex
D4	Infrastructure	SDNET2018	Local	Concrete cracks
D5	Logical Anomaly	MVTec-LOCO	Logical	Rule violations
D6	3D Product (RGB)	Real3D-AD	3D/Geom.	Shape deformations
D7	Remote Sensing	LEVIR-CD+	Change Det.	Building damage
D8	Dermatology	DermaMNIST	Medical	Subtle morphology
D9	Brain MRI	BraTS2021	Medical	Tissue overlap
D10	Liver CT	BMAD-Liver	Medical	Low-contrast lesions
D11	GI Endoscopy	HyperKvasir	Medical	Polyp detection
D12	Road Safety	BDD100K+RA21	Scene	Contextual reasoning
Total				1,440 images

Table 3: **Per-domain AUROC on the 12-domain benchmark.** Green = best; orange = second best. AnomalyClaw results use **[primary VLM]** with debate depth $D=2$.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	Macro
<i>Expert-only (no VLM)</i>													
CLIP-ZeroShot	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
WinCLIP	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
PatchCore Expert	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
<i>Single-pass VLM</i>													
VLM-Direct	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
Retrieval+VLM	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
Expert-Informed VLM	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
<i>Multi-round / agent</i>													
AgentIAD	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
Symmetric Debate	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX
AnomalyClaw (Ours)	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XX	bd.XXX

6. **Expert-Informed VLM:** Retrieval + expert evidence as text context, single-pass VLM judgment (AnomalyClaw without debate).

Multi-round/agent methods:

7. **AgentIAD** [Miao et al., 2025]: Tool-augmented single agent with iterative reasoning.
8. **Symmetric Debate:** Two VLM agents with identical roles debating anomaly presence (no Proposer/Advocate asymmetry, no expert grounding).

Implementation details. Unless otherwise noted, experiments use **[primary VLM]** as the debate backbone with temperature 0 and structured JSON output. The patch expert uses DINOv2 ViT-S/14 at native 518×518 resolution. Default debate depth is $D_{\max} = 2$. We report AUROC as the primary metric. Confidence intervals are 95% bootstrap CIs over 10,000 resamples.

4.2 Main Results

Table 3 presents per-domain AUROC for all methods.

Key observations. [Analysis of main results to be written after experiments complete. Expected observations: (1) AnomalyClaw should outperform all baselines on macro AUROC. (2) The gap over single-pass VLM methods should be larger than in the previous pipeline design, since debate catches hallucinations. (3) Expert-only baselines (CLIP-ZeroShot, WinCLIP) should be substantially weaker, establishing a clear progression. (4) Symmetric debate should underperform AnomalyClaw, validating the asymmetric design. (5) Largest gains expected on D5 (logical), D3 (multi-object), and medical domains where self-verification matters most.]

Table 4: **Debate ablation.** Adversarial debate with expert grounding outperforms all single-pass and symmetric alternatives.

Reasoning Strategy	Expert Evidence	Macro AUROC	VLM Calls/Query
Single-pass VLM	None	X.XXX	1
Single-pass VLM	Patch + Retrieval	X.XXX	1
Symmetric debate (no expert)	None	X.XXX	4
Symmetric debate	Patch + Retrieval	X.XXX	4
AnomalyClaw (asymmetric)	Patch + Retrieval	X.XXX	2–4

Table 5: **Expert pool ablation.** Patch-level evidence provides the largest single improvement; combining multiple experts yields additional gains.

Expert Configuration	Macro AUROC
No experts (debate only)	X.XXX
Retrieval expert only	X.XXX
Patch expert only	X.XXX
Patch + Retrieval	X.XXX
Patch + Retrieval + Texture	X.XXX

4.3 Ablation Studies

Ablation 1: Does adversarial debate help? Table 4 isolates the contribution of the debate mechanism by comparing AnomalyClaw variants with identical expert evidence but different reasoning strategies.

Ablation 2: Which experts matter? Table 5 tests the contribution of different experts within the AnomalyClaw framework.

Ablation 3: Debate depth. Table 6 varies the maximum debate depth $D_{\max}$.

4.4 Multi-VLM Generalization

A key claim of AnomalyClaw is VLM-agnostic design. Table 7 evaluates the framework across multiple VLM backends.

4.5 Analysis

Cost–accuracy tradeoff. The autonomous controller enables a natural cost–accuracy tradeoff through debate depth control. **[Pareto plot comparing: PatchCore Expert (0 VLM calls), single-pass VLM (1 call), AnomalyClaw D=1 (2 calls), AnomalyClaw D=2 (2–4 calls), symmetric debate (4 calls). AnomalyClaw should Pareto-dominate symmetric debate (better accuracy, similar or lower cost).]**

When does debate help most? **[Per-domain analysis showing debate gain over single-pass. Expected: largest gains on D5 (logical—debate catches compositional errors), D3 (multi-object—debate resolves ambiguous multi-region claims), and medical domains (debate catches hallucinated defects on normal tissue). Smallest gains on D10 (road safety—semantically obvious anomalies).]**

Autonomous controller behavior. **[Statistics on expert selection frequency and debate depth per domain. Expected: D10 mostly 1-round (easy), medical domains mostly 2-round (ambiguous), D5 frequently triggers all experts (complex).]**

Failure analysis. **[Identify worst-performing domain (likely D8 Liver CT) and analyze error modes. Focus on: Does debate reduce false positives? Does it help with false negatives? What types of anomalies resist adversarial verification?]**

Table 6: **Effect of debate depth.** Performance improves from $D=1$ to $D=2$ and plateaus at $D=3$.

$D_{\max}$	Macro AUROC	Avg. VLM Calls/Query
1	X.XXX	2.0
2	X.XXX	X.X
3	X.XXX	X.X
4	X.XXX	X.X

Table 7: **Cross-VLM generalization.** AnomalyClaw consistently improves over single-pass baselines regardless of the underlying VLM.

VLM Backend	Single-Pass AUROC	AnomalyClaw AUROC
VLM-A	X.XXX	X.XXX
VLM-B	X.XXX	X.XXX
VLM-C (open)	X.XXX	X.XXX

5 Conclusion

We presented AnomalyClaw, a training-free framework for cross-domain visual anomaly detection built on adversarial debate between an Anomaly Proposer and a Normality Advocate, grounded by an extensible pool of lightweight vision experts and orchestrated by an autonomous controller. The adversarial structure provides built-in self-verification—every anomaly claim must survive explicit counter-arguments before acceptance—addressing the hallucination and lack-of-verification problems that limit single-pass VLM approaches.

Evaluated on 1,440 test items across 12 diverse domains, AnomalyClaw achieves **X.XXX** macro AUROC, outperforming single-pass VLM baselines (**X.XXX**), expert-only methods (**X.XXX**), and symmetric multi-agent debate (**X.XXX**). The framework generalizes across multiple VLM backends without modification, and the autonomous controller adapts reasoning depth to query difficulty, reducing cost on easy cases while investing more computation on ambiguous ones.

Limitations. Performance on medical domains (particularly liver CT) remains below clinical requirements, reflecting fundamental limits of natural-image-pretrained features on medical imagery. The resolution thresholds (τ_{accept} , τ_{reject}) in the claim resolution rule are fixed; learning these from data could improve adaptability. Our benchmark evaluates 120 items per domain; larger-scale evaluation would strengthen statistical power. The autonomous controller currently uses heuristic rules for expert selection; a learned policy could further optimize the cost–accuracy tradeoff.

Future work. Extending the expert pool with domain-specific modules (CT windowing, frequency-domain analysis) could address medical domain weaknesses. Beyond classification, extending the debate to anomaly localization—where the Proposer specifies regions and the Advocate challenges spatial precision—is a natural next step. Integrating active learning, where uncertain debate outcomes trigger human review and update the system’s reference bank, represents a path toward continual improvement in deployed settings.

References

Ujjwal Baid, Satyam Ghodasara, Michel Bilello, Suyash Mohan, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C. Kitamura, Sarthak Pati, Luciano M. Prevedello, Jeffrey D. Rudie, Chiharu Sako, Russell T. Shinohara, Timothy Bergquist, Rong Chai, James A. Eddy, Julia Elliott, Walter Reade, Thomas Schaffter, Thomas Yu, Jiaxin Zheng, BraTS Annotators, Christos Davatzikos, John Mongan, Christopher Hess, Soonmee Cha, Javier E. Villanueva-Meyer, John B. Freymann, Justin S. Kirby, Benedikt Wiestler, Priscila Crivellaro, Rivka R. Colen, Aikaterini Kotrotsou, Daniel S. Marcus, Mikhail Milchenko, Arash Nazeri, Hassan M. Fathallah-Shaykh, Roland Wiest, András Jakab, Marc-André Weber, Abhishek Mahajan, Bjoern H. Menze, Adam E. Flanders, and Spyridon Bakas. The RSNA-ASNR-MICCAI brats 2021 benchmark on brain tumor segmentation and radiogenomic classification. *CoRR*, abs/2107.02314, 2021. URL <https://arxiv.org/abs/2107.02314>.

- Jinan Bao, Hanshi Sun, Hanqiu Deng, Yinsheng He, Zhaoxiang Zhang, and Xingyu Li. BMAD: benchmarks for medical anomaly detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024 - Workshops, Seattle, WA, USA, June 17-18, 2024*, pages 4042–4053. IEEE, 2024. doi: 10.1109/CVPRW63382.2024.00408. URL <https://doi.org/10.1109/CVPRW63382.2024.00408>.
- Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. Mvtec AD - A comprehensive real-world dataset for unsupervised anomaly detection. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pages 9592–9600. Computer Vision Foundation / IEEE, 2019. doi: 10.1109/CVPR.2019.00982. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Bergmann_MVTec_AD_-_A_Comprehensive_Real-World_Dataset_for_Unsupervised_Anomaly_CVPR_2019_paper.html.
- Paul Bergmann, Kilian Batzner, Michael Fauser, David Sattlegger, and Carsten Steger. Beyond dents and scratches: Logical constraints in unsupervised anomaly detection and localization. *Int. J. Comput. Vis.*, 130(4):947–969, 2022. doi: 10.1007/S11263-022-01578-9. URL <https://doi.org/10.1007/s11263-022-01578-9>.
- Hanna Borgli, Vajira Thambawita, Pia H. Smedsrud, Steven Hicks, Debesh Jha, Sigrun L. Eskeland, Kristin Ranheim Randel, Konstantin Pogorelov, Mathias Lux, Duc Tien Dang Nguyen, Dag Johansen, Carsten Griwodz, Håkon K. Stensland, Enrique Garcia-Ceja, Peter T. Schmidt, Hugo L. Hammer, Michael A. Riegler, Pål Halvorsen, and Thomas de Lange. Hyperkvasir, a comprehensive multi-class image and video dataset for gastrointestinal endoscopy. *Scientific Data*, 7(1):283, 2020. doi: 10.1038/s41597-020-00622-y. URL <https://doi.org/10.1038/s41597-020-00622-y>.
- Yunkang Cao et al. Anomaly-onevision: Towards unified visual anomaly detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025*, 2025. CVPR 2025.
- et al. Chen. M3-AD: Multi-modal multi-class anomaly detection. *CoRR*, 2025. arXiv preprint, 2025.
- Shuai Chen, Jinming Cao, Xin He, Zhihao Zhan, and Zhaoxiang Zhang. EAGLE: Expert-augmented visual prompting for training-free anomaly detection. *CoRR*, abs/2602.17419, 2026. URL <https://doi.org/10.48550/arXiv.2602.17419>.
- Sattar Dorafshan, Robert J. Thomas, and Marc Maguire. SDNET2018: An annotated image dataset for non-contact concrete crack detection using deep convolutional neural networks. *Data in Brief*, 21:1664–1668, 2018. doi: 10.1016/j.dib.2018.11.015. URL <https://doi.org/10.1016/j.dib.2018.11.015>.
- Zhaopeng Gu, Bingke Zhu, Guibo Zhu, Yingying Chen, Ming Tang, and Jinqiao Wang. Anomalygpt: Detecting industrial anomalies using large vision-language models. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Vancouver, Canada, February 20-27, 2024*, pages 1932–1940. AAAI Press, 2024. doi: 10.1609/AAAI.V38I3.27963. URL <https://doi.org/10.1609/aaai.v38i3.27963>.
- Jongheon Jeong, Yang Zou, Taewan Kim, Dongqing Zhang, Avinash Ravichandran, and Onkar Dabeer. Winclip: Zero-/few-shot anomaly classification and segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, pages 19606–19616. IEEE, 2023. doi: 10.1109/CVPR52729.2023.01878. URL <https://doi.org/10.1109/CVPR52729.2023.01878>.
- Dongwei Ji, Bingzhang Hu, and Yi Zhou. Autoiad: Manager-driven multi-agent collaboration for automated industrial anomaly detection. *CoRR*, abs/2508.05503, 2025. doi: 10.48550/ARXIV.2508.05503. URL <https://doi.org/10.48550/arXiv.2508.05503>.
- Xi Jiang, Jian Li, Hanqiu Deng, Yong Liu, Bin-Bin Gao, Yifeng Zhou, Jialin Li, Chengjie Wang, and Feng Zheng. MMAD: A comprehensive benchmark for multimodal large language models in industrial anomaly detection. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL <https://openreview.net/forum?id=JDIER86r8v>.

- et al. Li. FoundAD: Foundation model for anomaly detection. In *The Fourteenth International Conference on Learning Representations, ICLR 2026*. OpenReview.net, 2026.
- et al. Liang. M3MAD-Bench. *CoRR*, 2026. arXiv preprint, 2026.
- Hai Ling, Jia Guo, Zhulin Tao, Yunkang Cao, Donglin Di, Hongyan Xu, Xiu Su, Yang Song, and Lei Fan. Adnet: A large-scale and extensible multi-domain benchmark for anomaly detection across 380 real-world categories. *CoRR*, abs/2511.20169, 2025. doi: 10.48550/ARXIV.2511.20169. URL <https://doi.org/10.48550/arXiv.2511.20169>.
- Krzysztof Lis, Krishna Kanth Nakka, Pascal Fua, and Mathieu Salzmann. Detecting the unexpected via image resynthesis. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*, pages 2152–2161. IEEE, 2019. doi: 10.1109/ICCV.2019.00224. URL <https://doi.org/10.1109/ICCV.2019.00224>.
- Junwen Miao, Penghui Du, Yi Liu, Yu Wang, and Yan Wang. Agentiad: Tool-augmented single-agent for industrial anomaly detection. *CoRR*, abs/2512.13671, 2025. doi: 10.48550/ARXIV.2512.13671. URL <https://doi.org/10.48550/arXiv.2512.13671>.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision. *Trans. Mach. Learn. Res.*, 2024, 2024. URL <https://openreview.net/forum?id=a68SUt6zFt>.
- Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter V. Gehler. Towards total recall in industrial anomaly detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pages 14298–14308. IEEE, 2022. doi: 10.1109/CVPR52688.2022.01392. URL <https://doi.org/10.1109/CVPR52688.2022.01392>.
- Pinqiao Wang and Sheng Li. Multi-agent debate: A unified agentic framework for tabular anomaly detection. *CoRR*, abs/2602.14251, 2026. doi: 10.48550/ARXIV.2602.14251. URL <https://doi.org/10.48550/arXiv.2602.14251>.
- Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. Medmnist v2: A large-scale lightweight benchmark for 2d and 3d biomedical image classification. *Scientific Data*, 10(1):41, 2023. doi: 10.1038/s41597-022-01721-8. URL <https://doi.org/10.1038/s41597-022-01721-8>.
- Fisher Yu, Haofeng Chen, Xin Wang, Wenqi Xian, Yingying Chen, Fangchen Liu, Vashisht Madhavan, and Trevor Darrell. BDD100K: A diverse driving dataset for heterogeneous multitask learning. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*, pages 2633–2642. Computer Vision Foundation / IEEE, 2020. doi: 10.1109/CVPR42600.2020.00271. URL <https://doi.org/10.1109/CVPR42600.2020.00271>.
- et al. Zhang. AD-Copilot. *CoRR*, 2026. arXiv preprint, 2026.
- Jian Zhang, Runwei Ding, Miaoju Ban, and Linhui Dai. Pku-goodsad: A supermarket goods dataset for unsupervised anomaly detection and segmentation. *IEEE Robotics Autom. Lett.*, 9(3):2008–2015, 2024. doi: 10.1109/LRA.2024.3352358. URL <https://doi.org/10.1109/LRA.2024.3352358>.
- Zhehao Zhang, Ryan A. Rossi, Tong Yu, Franck Dernoncourt, Ruiyi Zhang, Jiuxiang Gu, Sungchul Kim, Xiang Chen, Zichao Wang, and Nedim Lipka. Vipact: Visual-perception enhancement via specialized VLM agent collaboration and tool-use. In *Fortieth AAAI Conference on Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, pages 36536–36546. AAAI Press, 2026. doi: 10.1609/AAAI.V40I43.40976. URL <https://doi.org/10.1609/aaai.v40i43.40976>.

Yang Zou, Jongheon Jeong, Latha Pemula, Dongqing Zhang, and Onkar Dabeer. Spot-the-difference self-supervised pre-training for anomaly detection and segmentation. In *Computer Vision - ECCV 2022 - 17th European Conference, Tel Aviv, Israel, October 23-27, 2022, Proceedings, Part XXX*, Lecture Notes in Computer Science, pages 392–408. Springer, 2022. doi: 10.1007/978-3-031-20056-4_23. URL https://doi.org/10.1007/978-3-031-20056-4_23.

A Additional Experimental Details

A.1 Benchmark Construction

Our benchmark draws from 12 established datasets, each contributing one domain. For each domain, we sample 60 normal and 60 anomalous images for the test split, totaling 1,440 test items. All sampling uses a fixed random seed (42) for reproducibility.

Domain Selection. Our initial data collection covered 14 domains. We excluded two: **Digital Forensics** because it requires detecting manipulation artifacts in compressed images, a signal-processing task rather than visual anomaly detection; and **Surveillance/Avenue** because it targets behavioral anomalies in video sequences, outside the scope of appearance-based single-image anomaly detection. The final benchmark comprises 12 domains encoded as D1–D12.

Source Datasets.

- **D1 (Industrial Manufacturing):** MVTec-AD [Bergmann et al., 2019]. 15 categories of manufactured products, sampled proportionally across all categories.
- **D2 (Retail Products):** GoodsAD [Zhang et al., 2024]. Consumer products with physical damage, opening defects, and surface damage.
- **D3 (Complex Industrial):** VisA [Zou et al., 2022]. Selected for its multi-object categories (PCBs, capsules) and complex texture objects (candles, cashews, macaroni), testing robustness beyond single-object inspection.
- **D4 (Infrastructure):** SDNET2018 [Dorafshan et al., 2018]. Concrete surface images (deck, pavement, wall) with crack annotations.
- **D5 (Logical Anomaly):** MVTec-LOCO [Bergmann et al., 2022]. Correctly assembled products vs. logical errors (wrong count, wrong position, wrong component type).
- **D6 (3D Product Inspection):** Real3D-AD. Rendered RGB views of 3D point cloud scans of industrial products. Normal products vs. geometric deformations (bulges, sinks, contamination). Tests whether VLMs can detect shape anomalies from rendered 3D data.
- **D7 (Remote Sensing):** LEVIR-CD+. Satellite imagery of buildings before and after change events. No-change (normal) vs. building construction, demolition, or damage. Tests cross-modality generalization to aerial imagery.
- **D8 (Dermatology):** DermaMNIST [Yang et al., 2023] and ISIC dermoscopic images. Melanocytic nevi (normal) vs. melanoma and other malignancies.
- **D9 (Brain MRI):** BraTS2021 [Baid et al., 2021]. Axial FLAIR slices. Normal slices vs. slices containing glioma.
- **D10 (Liver CT):** BMAD-Liver [Bao et al., 2024]. Axial CT slices of normal liver vs. focal lesions.
- **D11 (GI Endoscopy):** HyperKvasir [Borgli et al., 2020]. Normal colon mucosa vs. polyps and other pathology.
- **D12 (Road Safety):** BDD100K [Yu et al., 2020] (normal driving scenes) and Road-Anomaly21 [Lis et al., 2019] (scenes with unexpected obstacles).

Reference Images. Each test item is paired with up to $k=4$ normal reference images retrieved using DINOv2 visual similarity from a pre-built per-domain index.

A.2 Domain Knowledge Specifications

Each domain is equipped with a structured knowledge specification provided to the VLM agents. The specification contains: (1) a domain description, (2) what normal samples look like, (3) specific anomaly criteria, and (4) common false positives. An example for D1:

Domain: Industrial Manufacturing (MVTec-AD).

Normal: Defect-free manufactured products.

Anomaly criteria: Surface defects (scratches, stains, discoloration); structural defects

(cracks, breaks, missing parts); contamination (foreign objects, residue).

Common false positives: Lighting variation, slight rotation, normal manufacturing tolerance.

A.3 Debate Prompt Templates

Anomaly Proposer (cold start). The Proposer receives: (1) k normal reference images, (2) the query image, (3) domain knowledge text, and (4) expert evidence reports. It outputs a JSON object containing a `normal_profile` (structural description of what the object should look like) and a `claims` array. Each claim includes: `id`, `category`, `location` (bounding box or quadrant), `appearance`, `evidence` (linking to expert findings), `analysis`, `confidence` (0–1), and `severity`.

Normality Advocate. The Advocate receives the same images plus the Proposer’s claims. For each claim, it outputs: `refute_confidence` (0–1, how confident it is the observation is NOT anomalous), `counter_evidence` (specific reasons), and `likely_cause` (benign attribution such as lighting, texture variation, compression artifacts).

Refinement rounds. In subsequent rounds, both agents receive only the TBD claims from the previous round along with the opposing agent’s arguments. The Proposer may revise confidence levels; the Advocate may strengthen or withdraw challenges.

A.4 Patch Expert Implementation

Feature Extraction. DINOv2 ViT-S/14 (`vit_small_patch14_dinov2.lvd142m`) from `timm`. Input: 518×518 pixels, producing $37 \times 37 = 1,369$ patches of 384 dimensions. All features are L2-normalized.

Anomaly Score. For each query patch, cosine distance to nearest reference patch (Eq. 1). Anomaly score = mean of top 1% distances (≈ 14 patches).

Text Report Generation. The report includes: (1) raw mean patch distance with annotation “higher = more different”; (2) global CLS-token similarity; (3) qualitative assessment from sigmoid calibration $\sigma(20 \cdot (d - 0.18))$ binned into five levels.

A.5 Bootstrap Confidence Intervals

All confidence intervals: 10,000 bootstrap resamples, fixed seed. 95% CIs computed over 10 domain-level AUROCs.

[Bootstrap CI table to be populated after experiments.]

A.6 Per-Category Breakdown

[Per-category AUROC for D1 (MVTec-AD, 15 categories) and D3 (VisA, multi-object and complex categories) to verify gains are not driven by a subset.]

A.7 Detailed Failure Analysis

[Detailed error analysis for worst-performing domain, including patch distance distributions, debate transcript examples, and VLM error patterns.]